

rd_data toggles
and matches
stored values

MISO_Enable &
MISO asserted
during read cycles

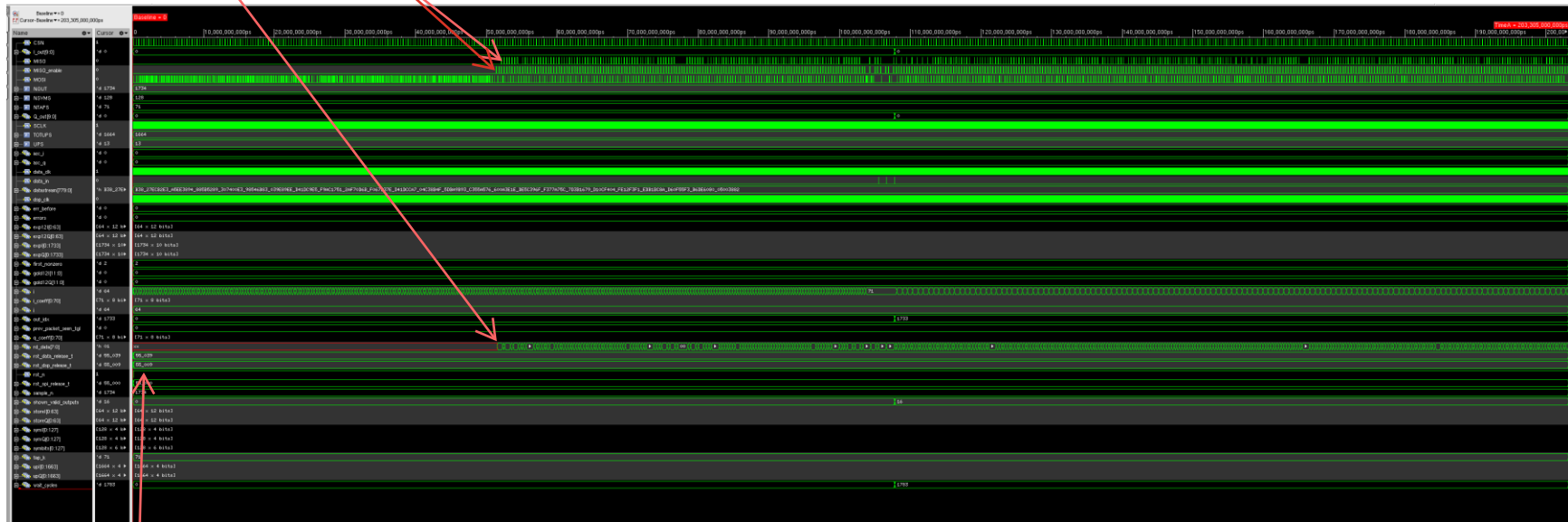


Figure 1: Waveform showing qam64_modulator_p9 with RTL simulation

rst_n internal
synchronized
resets

Sent data Packet

I_{out}/Q_{out} show
filtered waveform.
Verified I/Q outputs
for entire packet

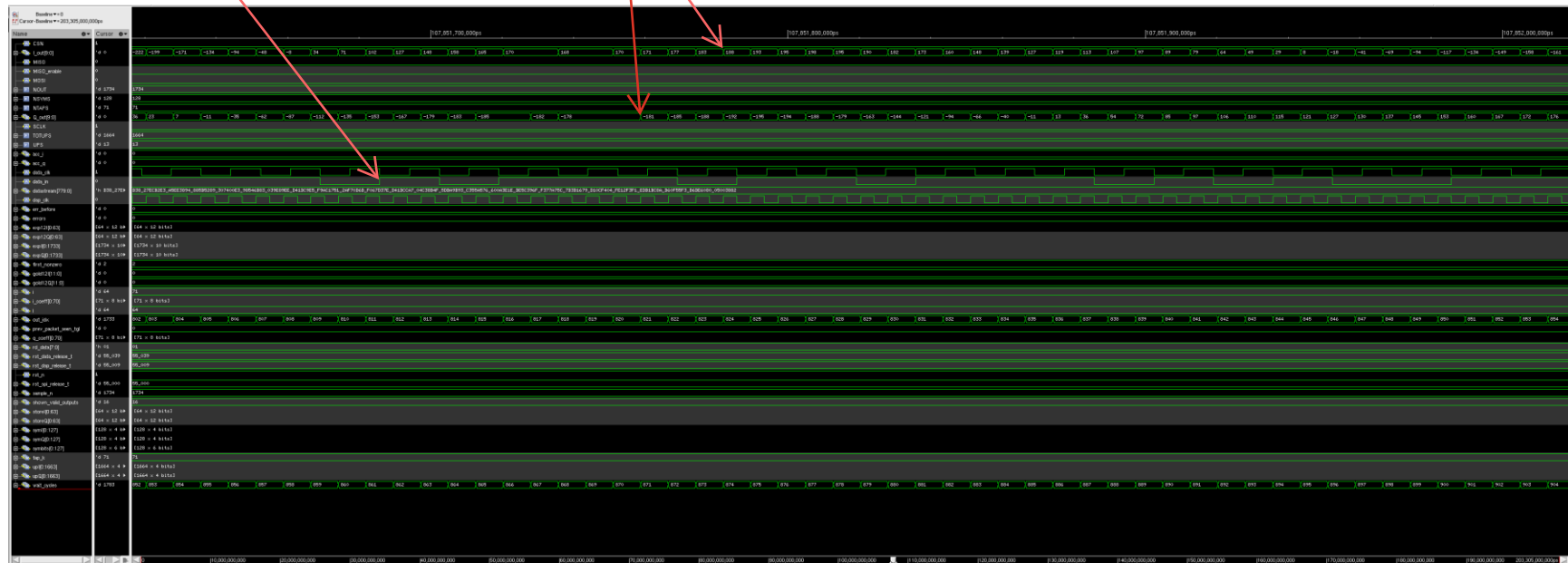


Figure 2: zoomed in Waveform showing qam64_modulator_p9 with RTL simulation

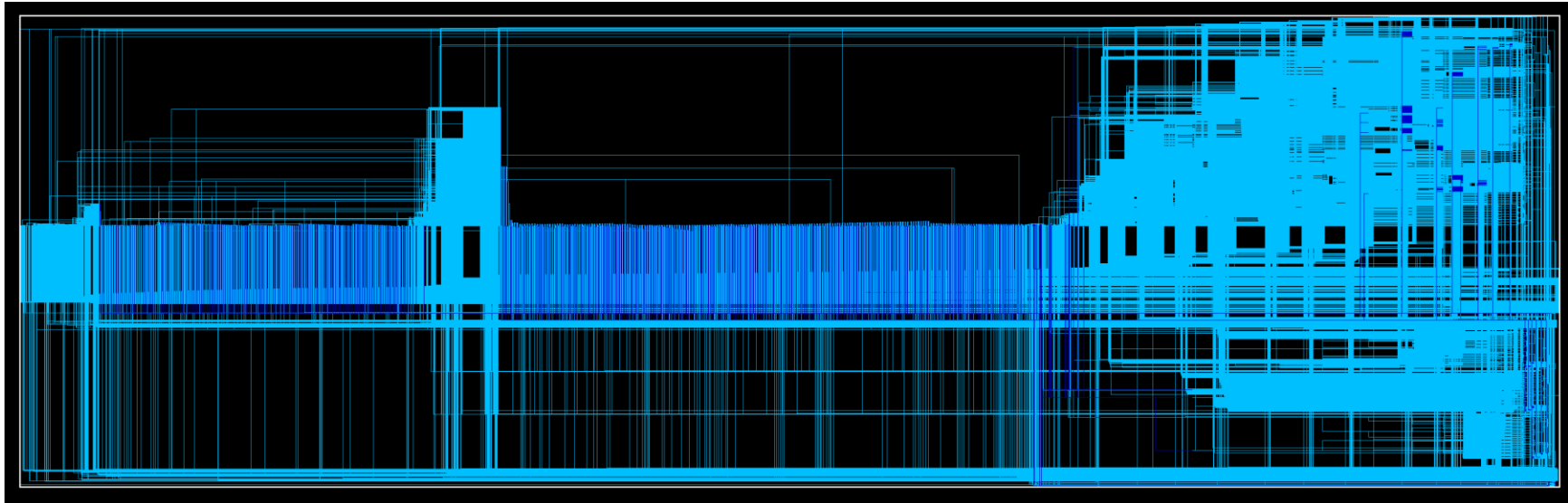


Figure 4: Synthesis schematic for qam64_modulator_p9_fixed

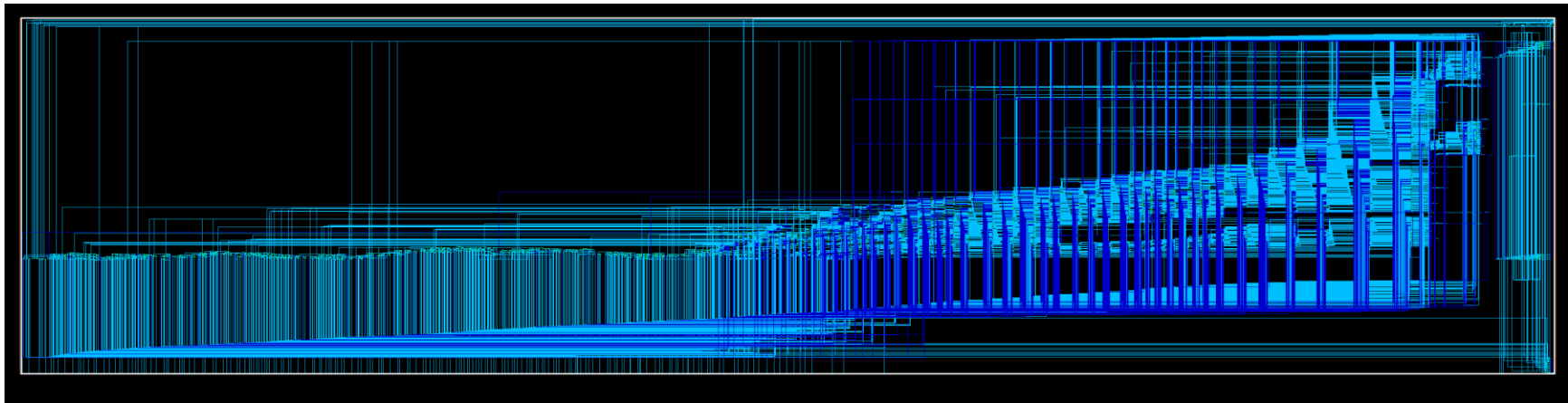


Figure 5: Synthesis Schematic for u_fir_q

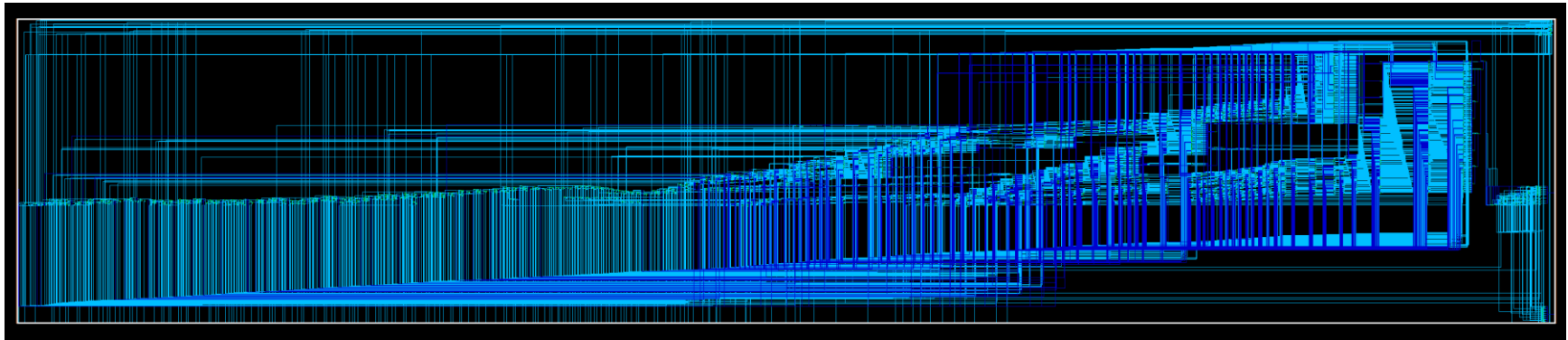


Figure 6: Synthesis Schematic for u_fir_i

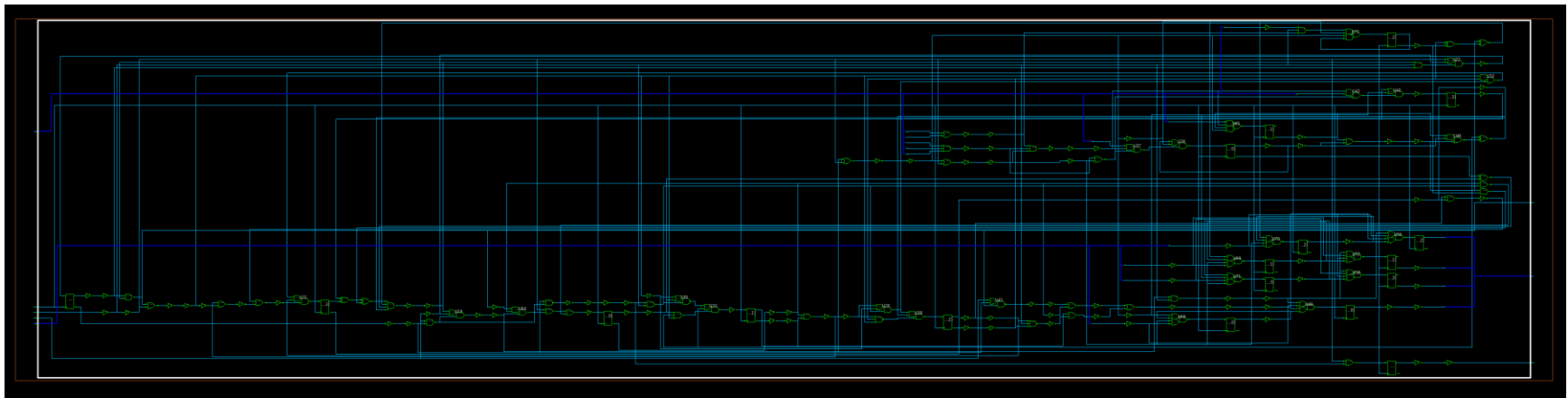


Figure 7: Synthesis Schematic for u_up_q

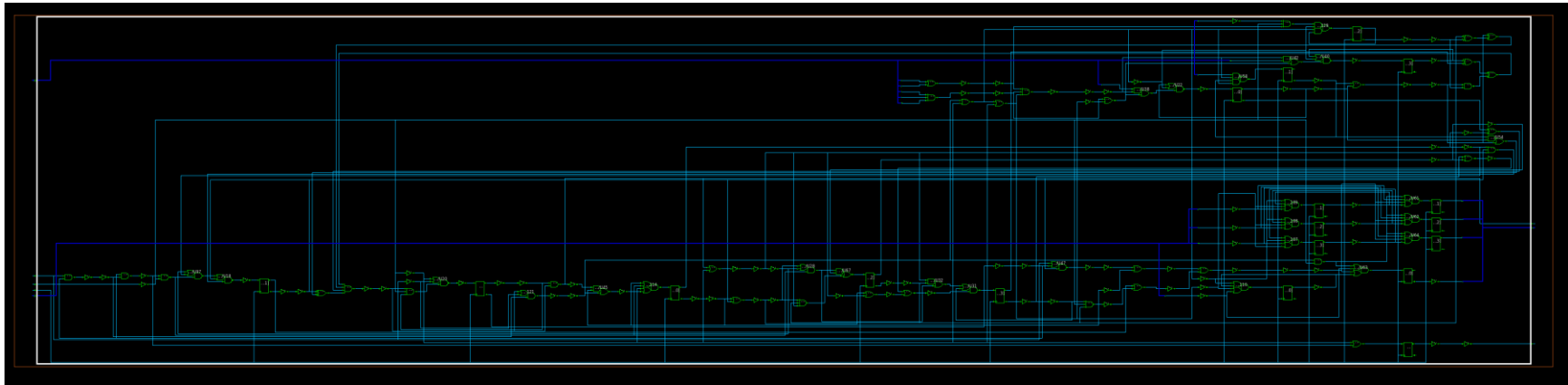


Figure 8: Synthesis Schematic for u_up_i

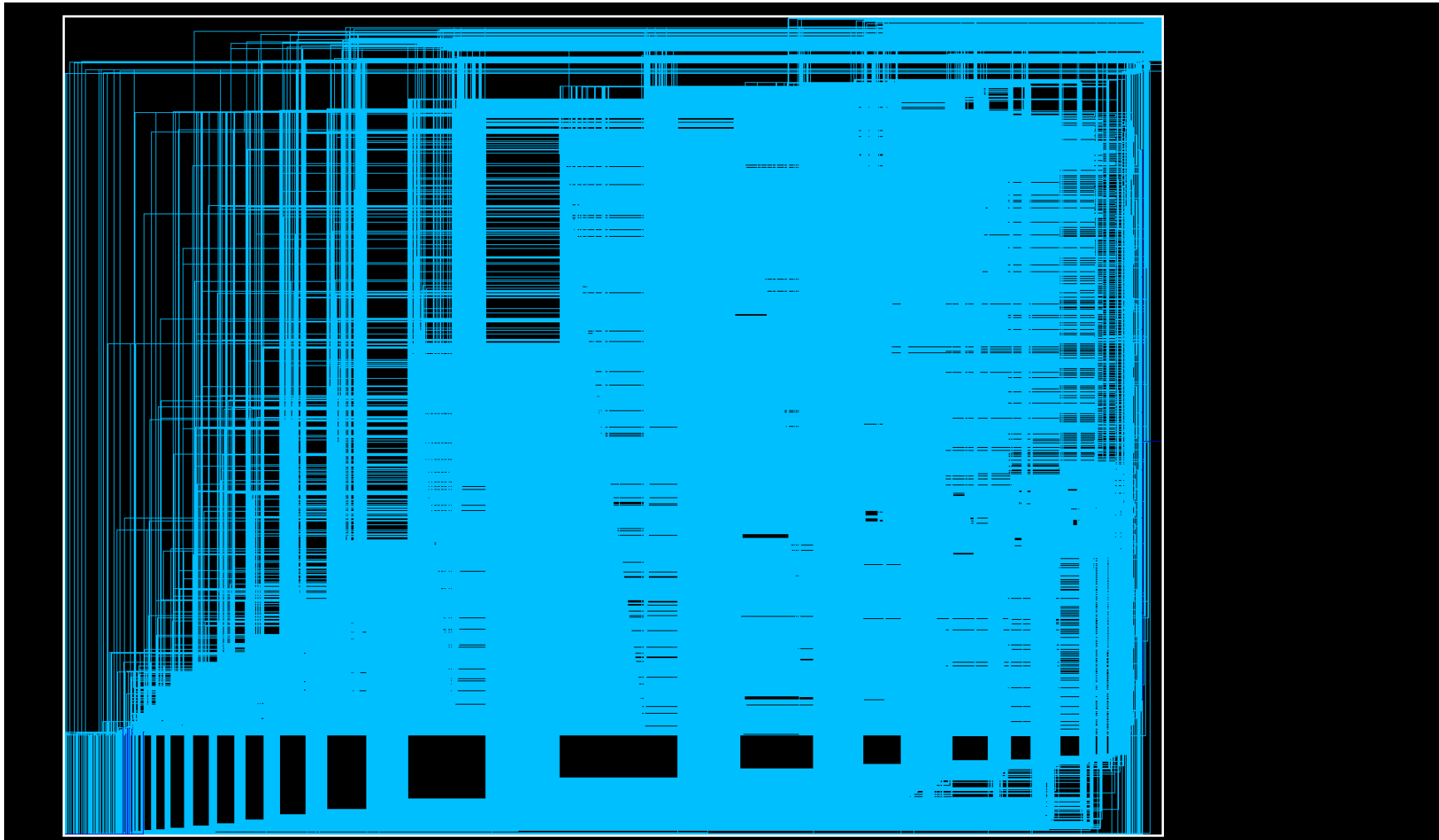


Figure 9: Synthesis Schematic for u_fifo

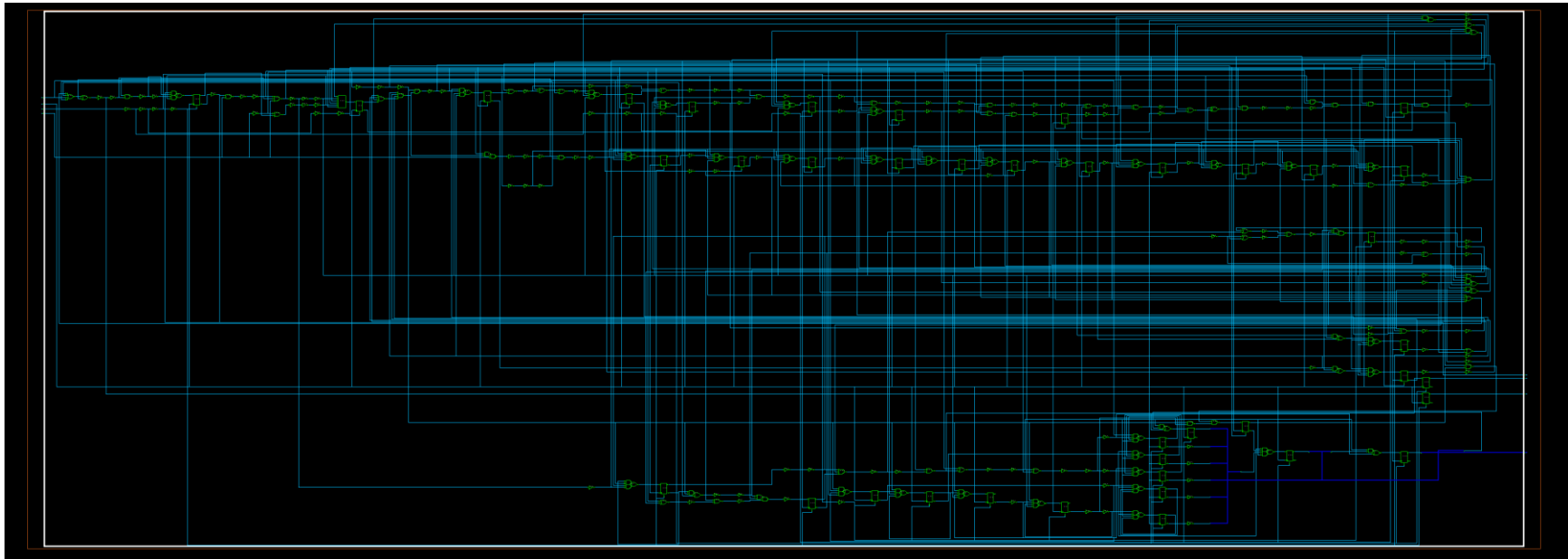


Figure 10: Synthesis Schematic for u_mapper

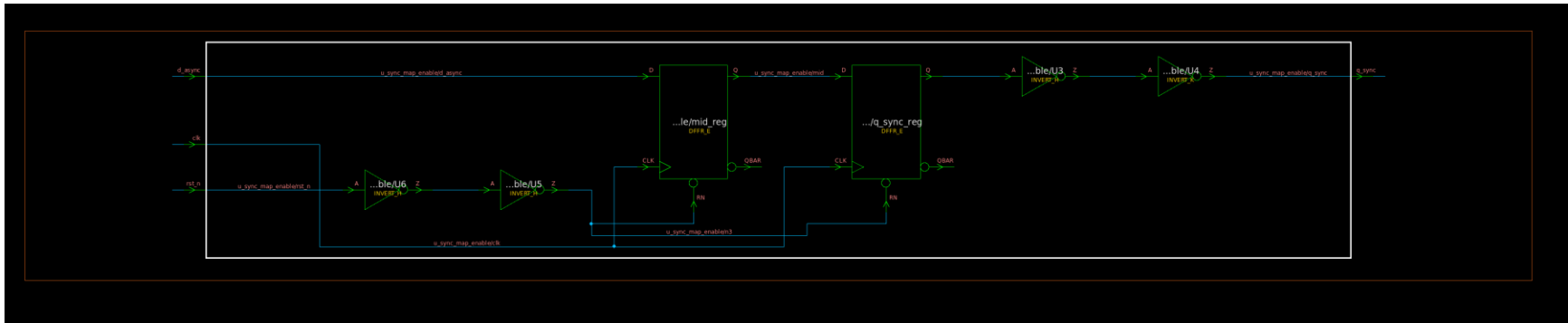


Figure 11: Synthesis Schematic for u_sync_map_enable

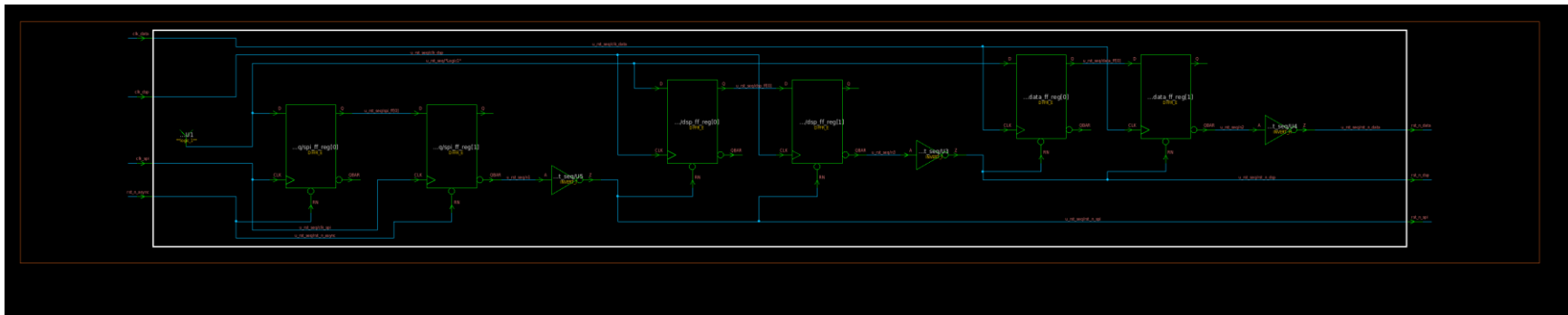


Figure 12: Synthesis Schematic for u_rst_seq

Constraints file is in zip folder

1. Max Frequency

- SCLK target: 100 kHz $\rightarrow$ Met, approx. max ~ 181.7 kHz from the reported margin on the shown half-cycle path.
- data_clk target: 60 MHz $\rightarrow$ Met, estimated $F_{max} \approx 273.45$ MHz.
- dsp_clk target: 130 MHz $\rightarrow$ Met, estimated $F_{max} \approx 130.0$ MHz.

2. Total Area: $1,323,114.250161 \mu m^2$ (From Area Report)

Step3: Back Annotated Simulations

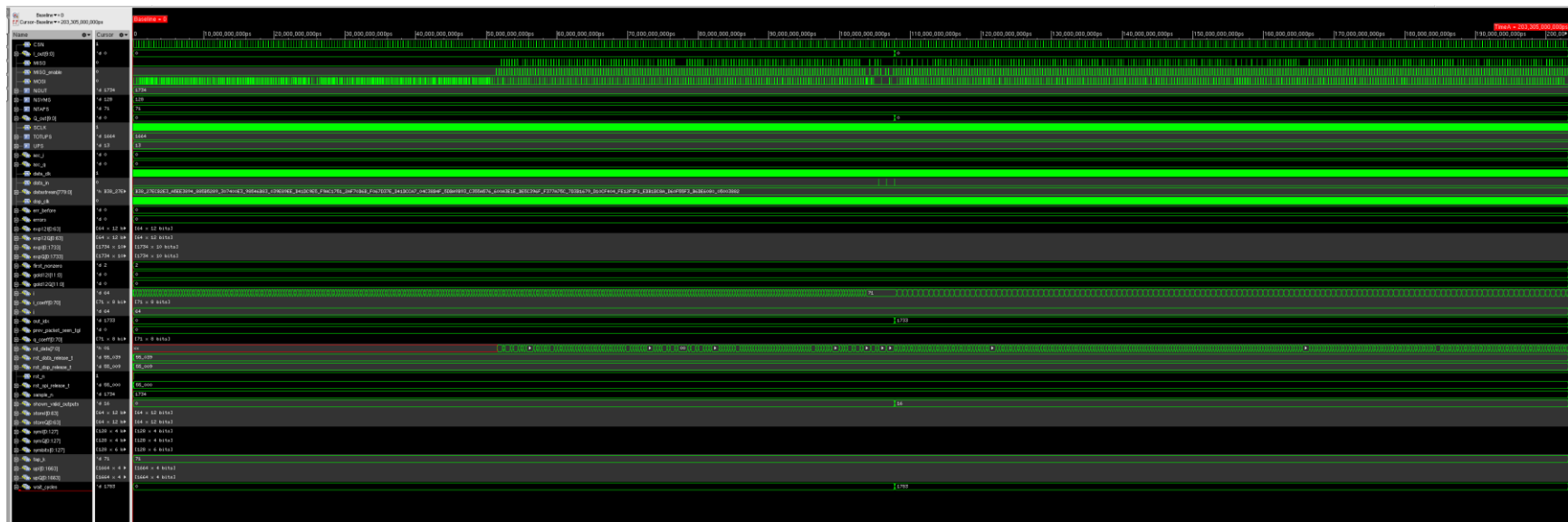


Figure 7: Waveform showing qam64_modulator_p9 with Back Annotated simulation

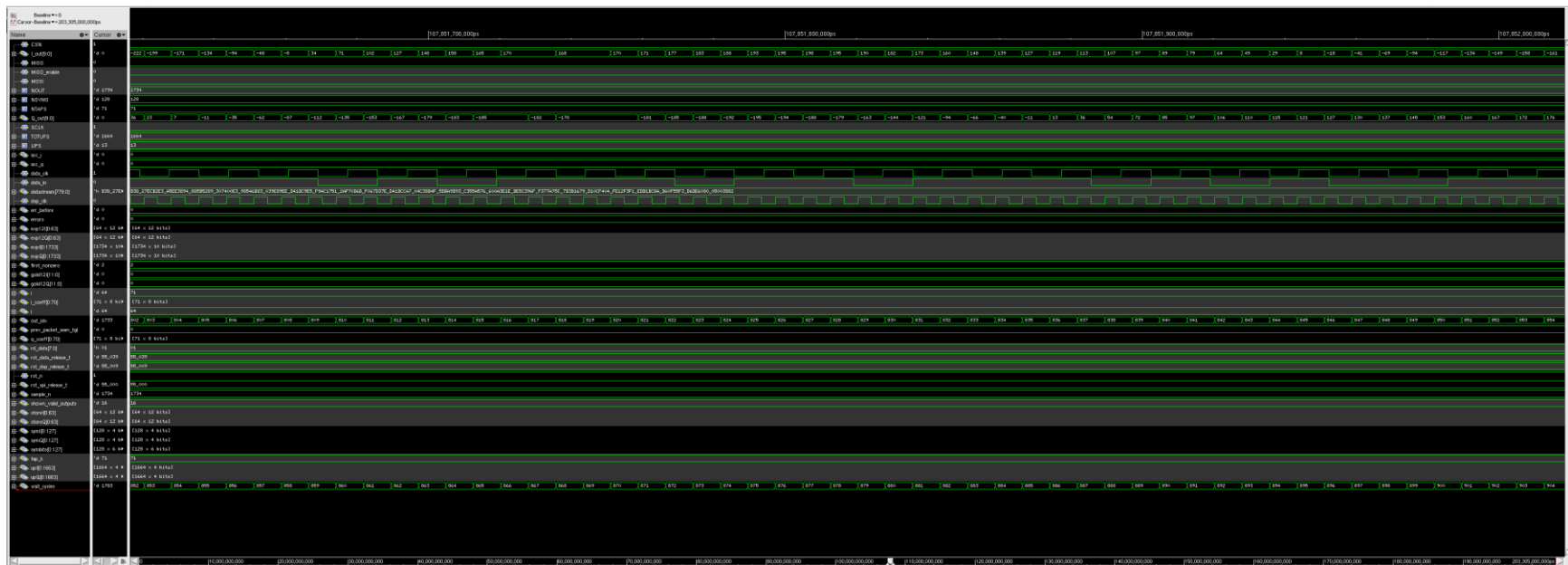
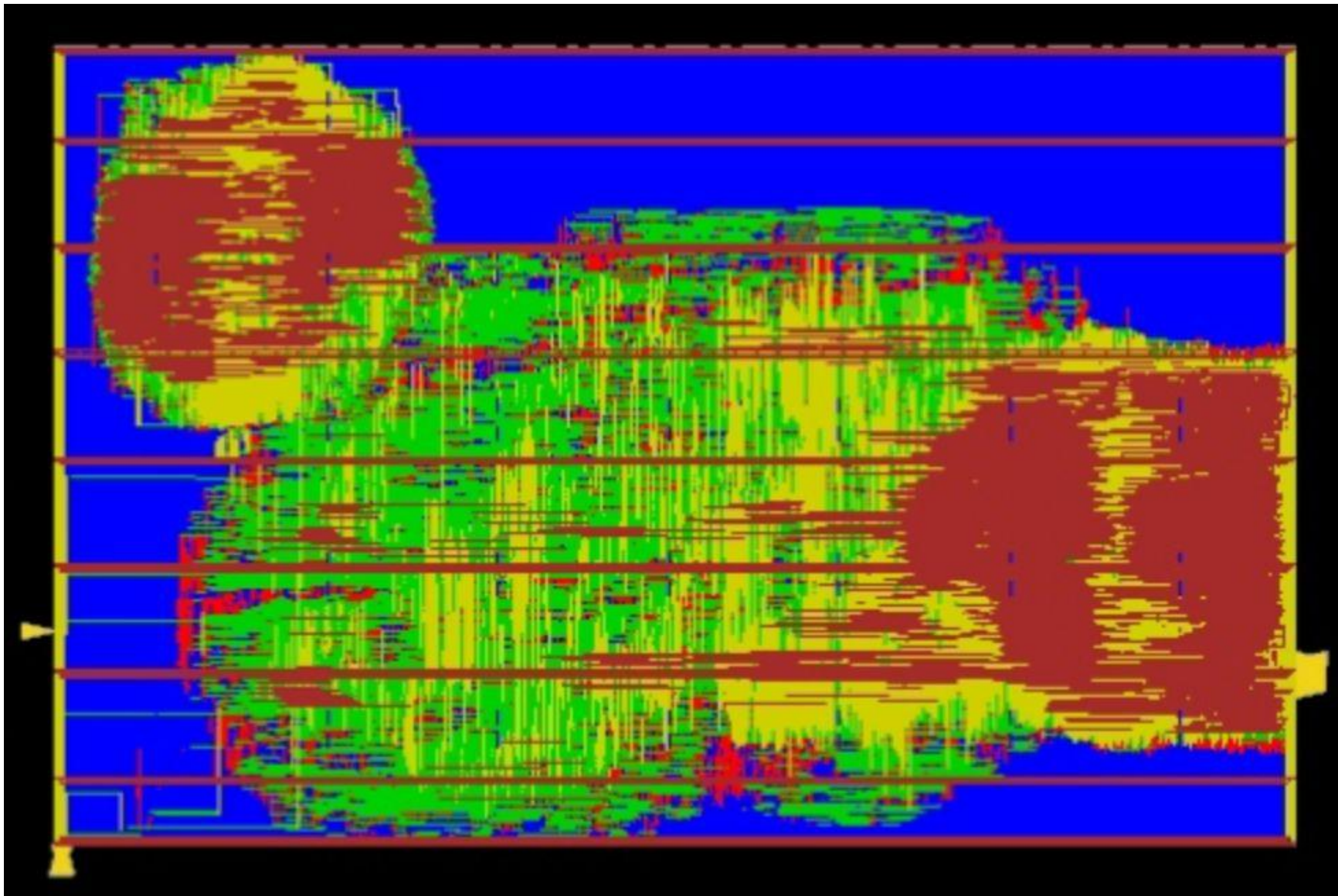


Figure 8: Zoomed in Waveform showing qam64_modulator with Back Annotated simulation

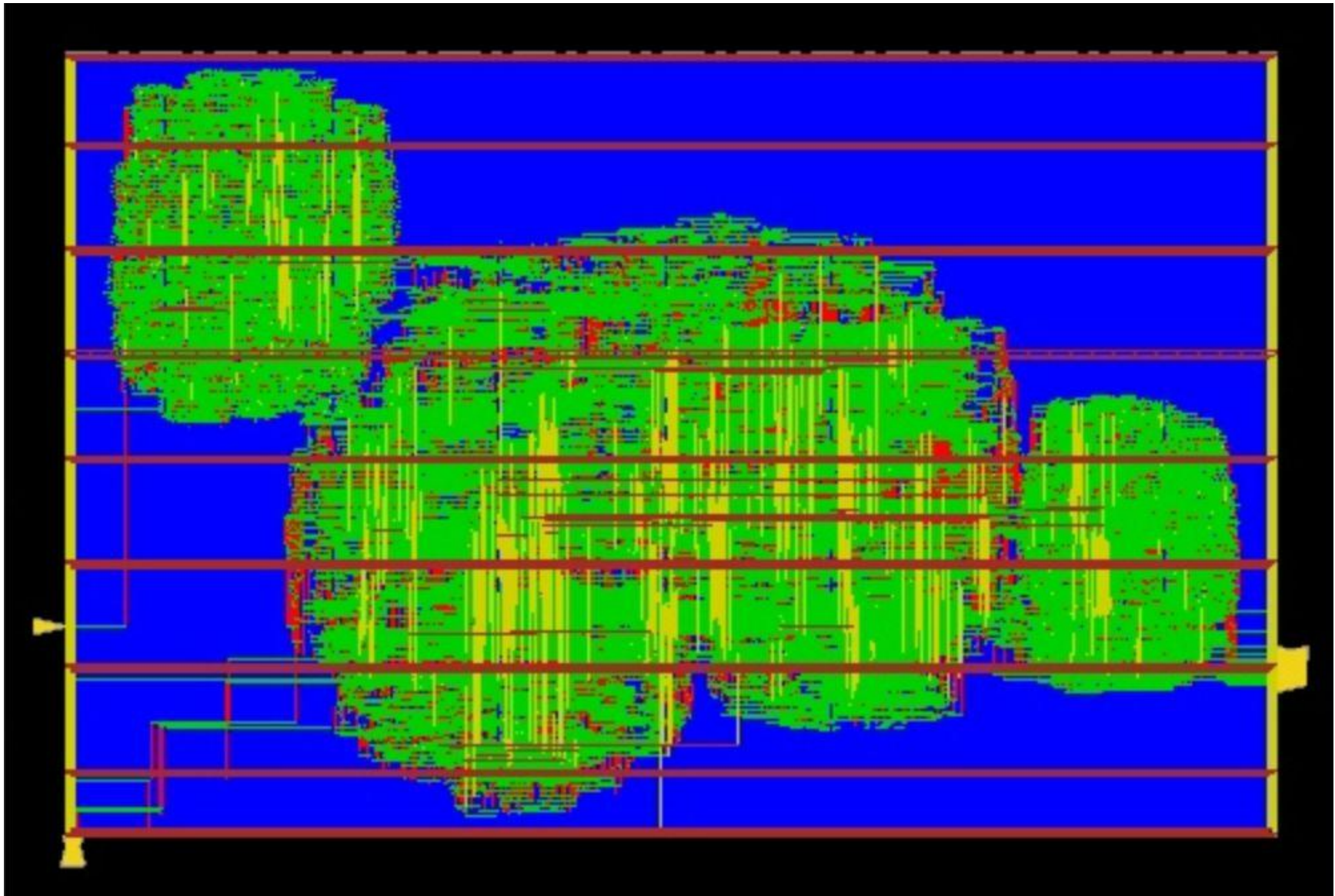
Layout

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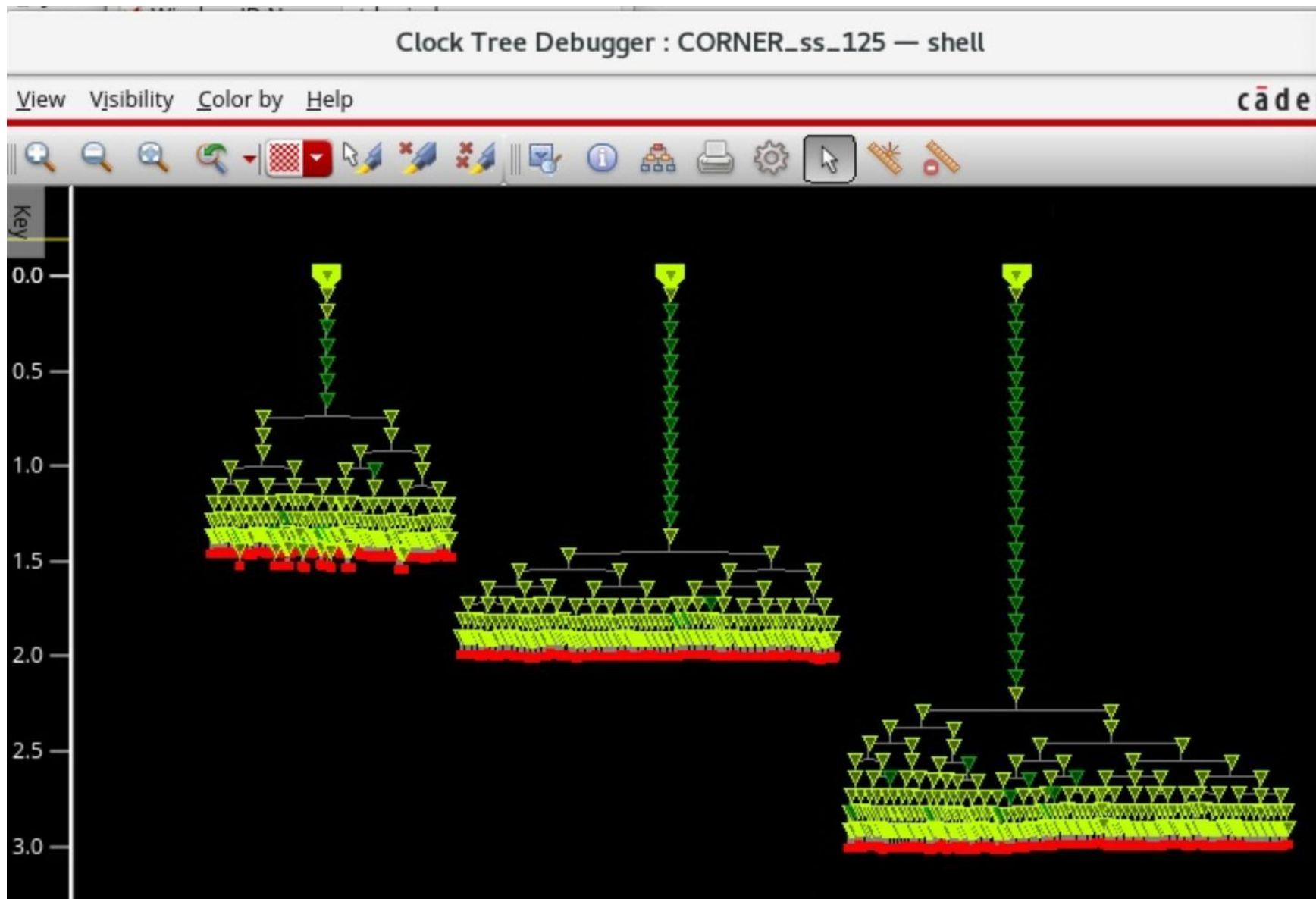
Layout Final

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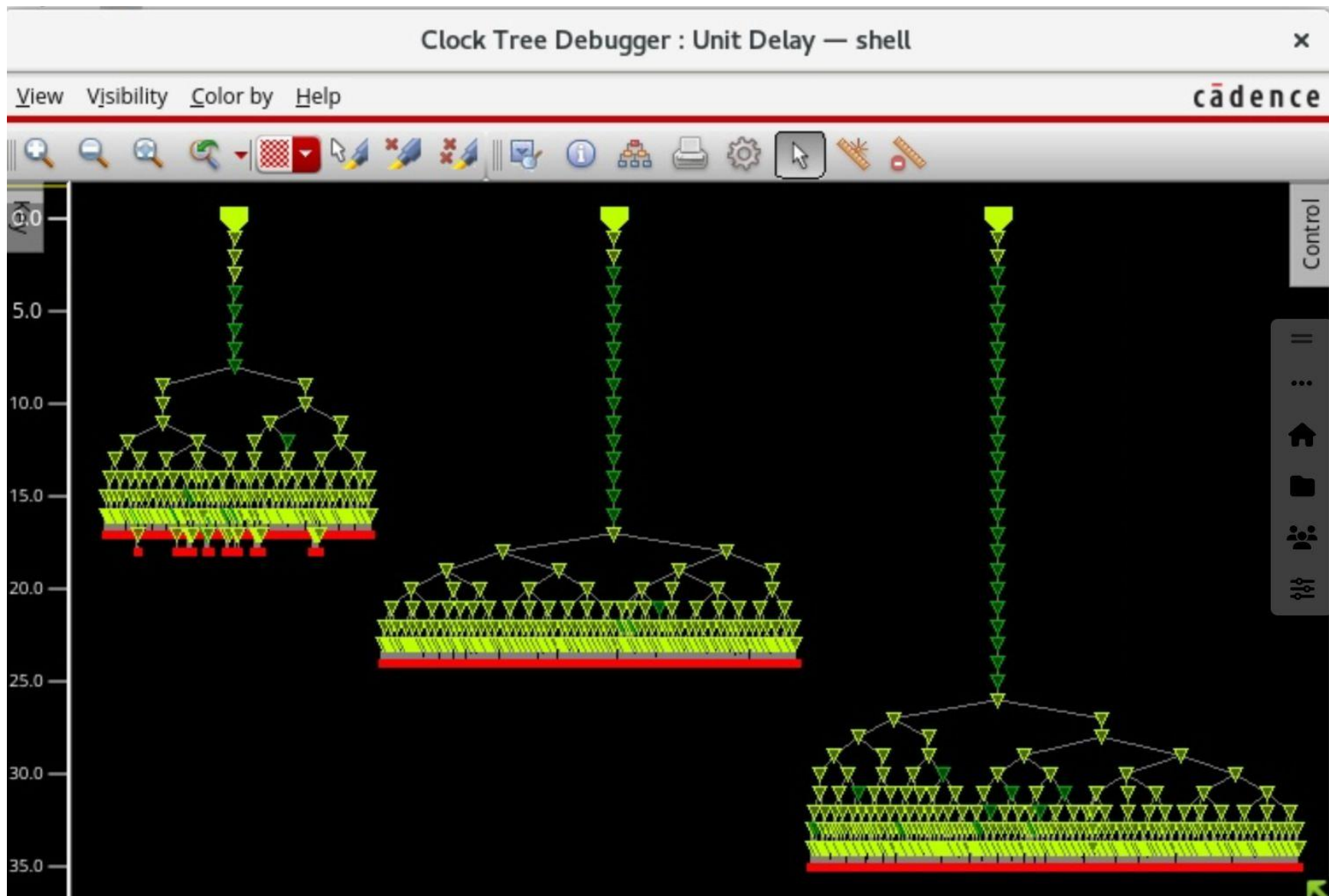
Floorplan

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Post_cts

SUBMITTED BY JOYCE BERDKAN (G30453881) AND SYED HASAN LNU (G43129353)



Original Clock Tree